TechBridge Data Analytics Internship — Task 8

Retail Sales Performance Analysis

Project Objective

To use the raw data to understand the sales performance and customer activity of the company; cleaning the data, combining the related tables, analysing the sales information, creating visualizations, and using SQL to answer business questions.

Dataset Description

There are three datasets: Customers, Products, and Transactions.
The Customers dataset has three columns: CustomerID, City, and CustomerType, and 20 rows.
The Products dataset has four columns: ProductID, ProductName, Category, and UnitPrice, and 10 rows.
The Transactions dataset has six columns: TransactionID, CustomerID, ProductID, TransactionDate, Quantity, and Discount, and 150 rows.
The three datasets are connected together with three of the columns: CustomerID and ProductID.
Important fields for our analysis are CustomerID, CustomerName, CustomerType, City, ProductID, ProductName, TransactionDate, UnitPrice, and Quantity.

Data Cleaning

- In the transaction table, there is an unmatched CustomerID (C999) which isn't in the Customer table. I deleted the row with the customerID C999, because the ID doesn't exist in the Customer table; it is an error entry.
- There are no duplicate rows found in each dataset table.
- There were no missing values or blank cells.
- Used the TRIM function to remove spaces.
- In the transaction table, there is an unmatched ProductID (P999) which isn't in the Product table. I deleted the row with the ProductID P999, because the ID doesn't exist in the Product table; it is an error entry.
- Corrected cell format and data types to ensure consistency and accurate analysis.

- In the Transaction table, in the Quantity field there is **-2** value for transaction T0120. This could imply that 2 items were ordered but not delivered to the customer, or that two items were delivered but the prices were not paid for by the client. **I have decided to flag it as an invalid record; for management to review. But I have to remove the row because it would affect my final analysis, especially the product category (giving a** -900,000.00 **value for laptop sold for** T0120**).**

BUSINESS ANALYSIS

Question 1 — Total Revenue

Business question:

> What is the total revenue generated from all transactions?

Answer: **#42,013,200.00**

Question 2 — Total Transactions

Business question:

> How many transactions were recorded?

Answer: **147** transactions.

Question 3 — Total Quantity Sold

Business question:

> How many units of products were sold?

Answer: **447** units of products sold.

Question 4 — Best-Performing Product

Business question:

> Which product generated the highest revenue?

Answer: **Monitor** has the highest revenue with **#7,992,000.00**

Best-Performing Product

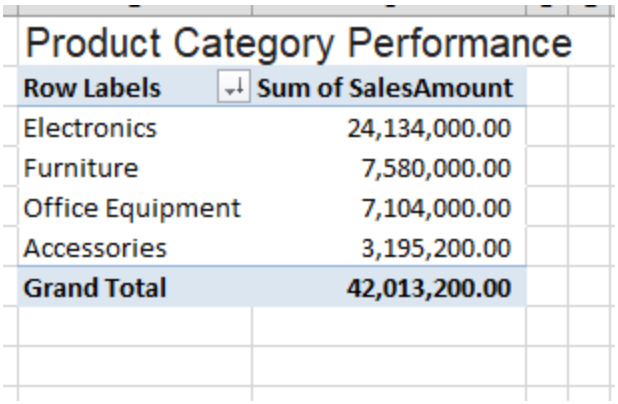
Row Labels	Sum of SalesAmount
Monitor	7,992,000.00
Printer	7,104,000.00
Tablet	6,512,000.00
Desk	6,174,000.00
Laptop	5,220,000.00
External Drive	4,410,000.00
Headset	1,850,000.00
Office Chair	1,406,000.00
Keyboard	1,110,000.00
Mouse	235,200.00
Grand Total	**42,013,200.00**

Question 5 — Product Category Performance

Business question:

> Which product category generated the highest revenue?

Answer: **Electronics** has the highest performance at **#24,134,000.00** **Accessories** has the lowest performance at **#3,195,200.00**



Product Category Performance

Row Labels	Sum of SalesAmount
Electronics	24,134,000.00
Furniture	7,580,000.00
Office Equipment	7,104,000.00
Accessories	3,195,200.00
Grand Total	**42,013,200.00**

Question 6 — Customer Performance

Business question:

> Which customers generated the highest revenue?

Answer: **Ibrahim Sani** has the highest generated revenue at **#4,716,000.00**

Customer Performance	
Row Labels	**Sum of SalesAmount**
Ibrahim Sani	4,716,000.00
Hauwa Ali	3,768,000.00
Tunde Adebayo	3,726,000.00
Mariam Yusuf	3,454,000.00
Fatima Musa	3,336,000.00
Ruth Daniel	3,297,000.00
Esther James	3,082,500.00
Joy Akpan	3,058,000.00
Grace Eze	2,877,000.00
Daniel Okafor	2,355,000.00
David Umeh	2,055,000.00
Aisha Bello	1,687,500.00
Chinedu Obi	993,750.00
Michael Joseph	856,250.00
Peter Williams	755,250.00
Samuel Adeyemi	650,750.00
Blessing Ojo	588,750.00
Mercy John	521,250.00
Emeka Nwosu	125,600.00
Kelvin George	109,600.00
Grand Total	**42,013,200.00**

Question 7 — Location Performance

Business question:

> Which city generated the highest revenue?

Answer: **Kano** has the highest location performance at **#7,222,000.00**

Location Performance	
Row Labels	**Sum of SalesAmount**
Kano	7,222,000.00
Abuja	7,071,000.00
Lagos	5,573,250.00
Ibadan	4,376,750.00
Benin City	3,398,250.00
Kaduna	3,336,000.00
Warri	3,082,500.00
Uyo	3,058,000.00
Asaba	2,055,000.00
Port Harcourt	1,103,350.00
Calabar	856,250.00
Abeokuta	755,250.00
Enugu	125,600.00
Grand Total	**42,013,200.00**

Question 8 — Customer Type

Business question:

> Which customer type generates the most revenue?

Answer: **Retail** customers have the highest revenue at **#23,075,850.00**

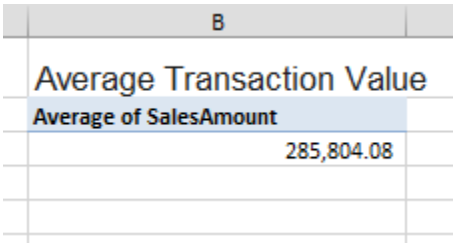
Customer Type Performance	
Row Labels	**Sum of SalesAmount**
Retail	23,075,850.00
Corporate	16,320,350.00
Wholesale	2,617,000.00
Grand Total	**42,013,200.00**

Question 9 — Average Transaction Value

Business question:

> What is the average revenue generated per transaction?

Answer: **#285,804.08** is the average revenue generated per customer purchase.



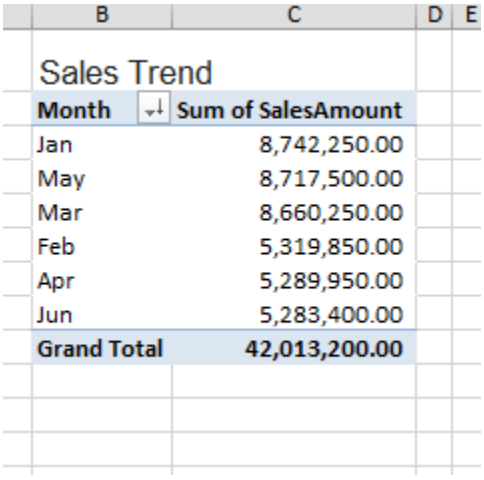
	B
	Average Transaction Value
	Average of SalesAmount
	285,804.08

Question 10 — Sales Trend

Business question:

> How did revenue change over time?

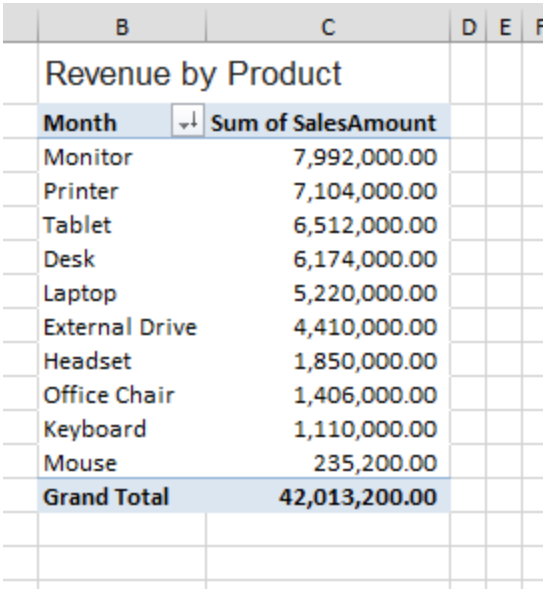
Answer: **January** has the highest-sales month, while **June** has the lowest-sales month. The trend shows that sales generally were **fluctuating**.



	B	C	D	E
	Sales Trend			
	Month	Sum of SalesAmount		
	Jan	8,742,250.00		
	May	8,717,500.00		
	Mar	8,660,250.00		
	Feb	5,319,850.00		
	Apr	5,289,950.00		
	Jun	5,283,400.00		
	Grand Total	42,013,200.00		

PIVOT TABLES

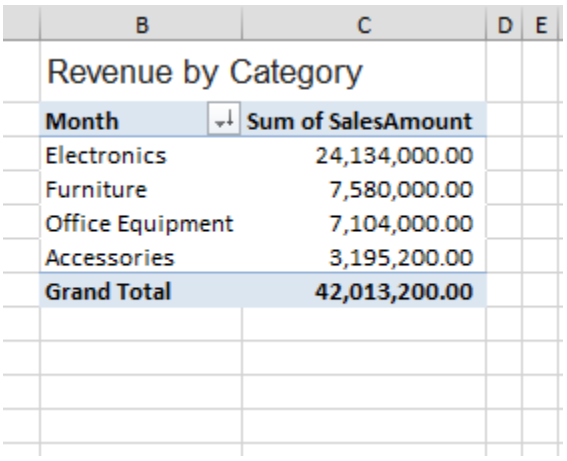
Pivot Table 1 — Revenue by Product



Revenue by Product

Month	Sum of SalesAmount
Monitor	7,992,000.00
Printer	7,104,000.00
Tablet	6,512,000.00
Desk	6,174,000.00
Laptop	5,220,000.00
External Drive	4,410,000.00
Headset	1,850,000.00
Office Chair	1,406,000.00
Keyboard	1,110,000.00
Mouse	235,200.00
Grand Total	**42,013,200.00**

Pivot Table 2 — Revenue by Category



Revenue by Category

Month	Sum of SalesAmount
Electronics	24,134,000.00
Furniture	7,580,000.00
Office Equipment	7,104,000.00
Accessories	3,195,200.00
Grand Total	**42,013,200.00**

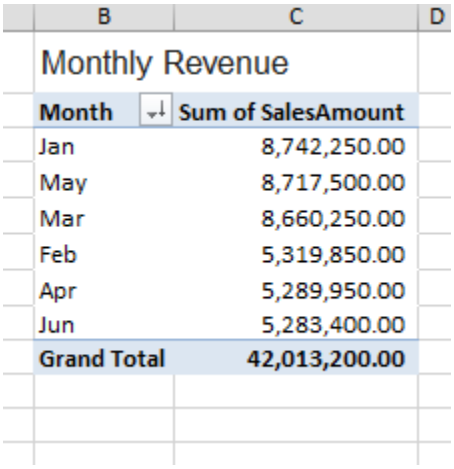
Pivot Table 3 — Revenue by City

Revenue by City	
Month	**Sum of SalesAmount**
Kano	7,222,000.00
Abuja	7,071,000.00
Lagos	5,573,250.00
Ibadan	4,376,750.00
Benin City	3,398,250.00
Kaduna	3,336,000.00
Warri	3,082,500.00
Uyo	3,058,000.00
Asaba	2,055,000.00
Port Harcourt	1,103,350.00
Calabar	856,250.00
Abeokuta	755,250.00
Enugu	125,600.00
Grand Total	**42,013,200.00**

Pivot Table 4 — Revenue by Customer

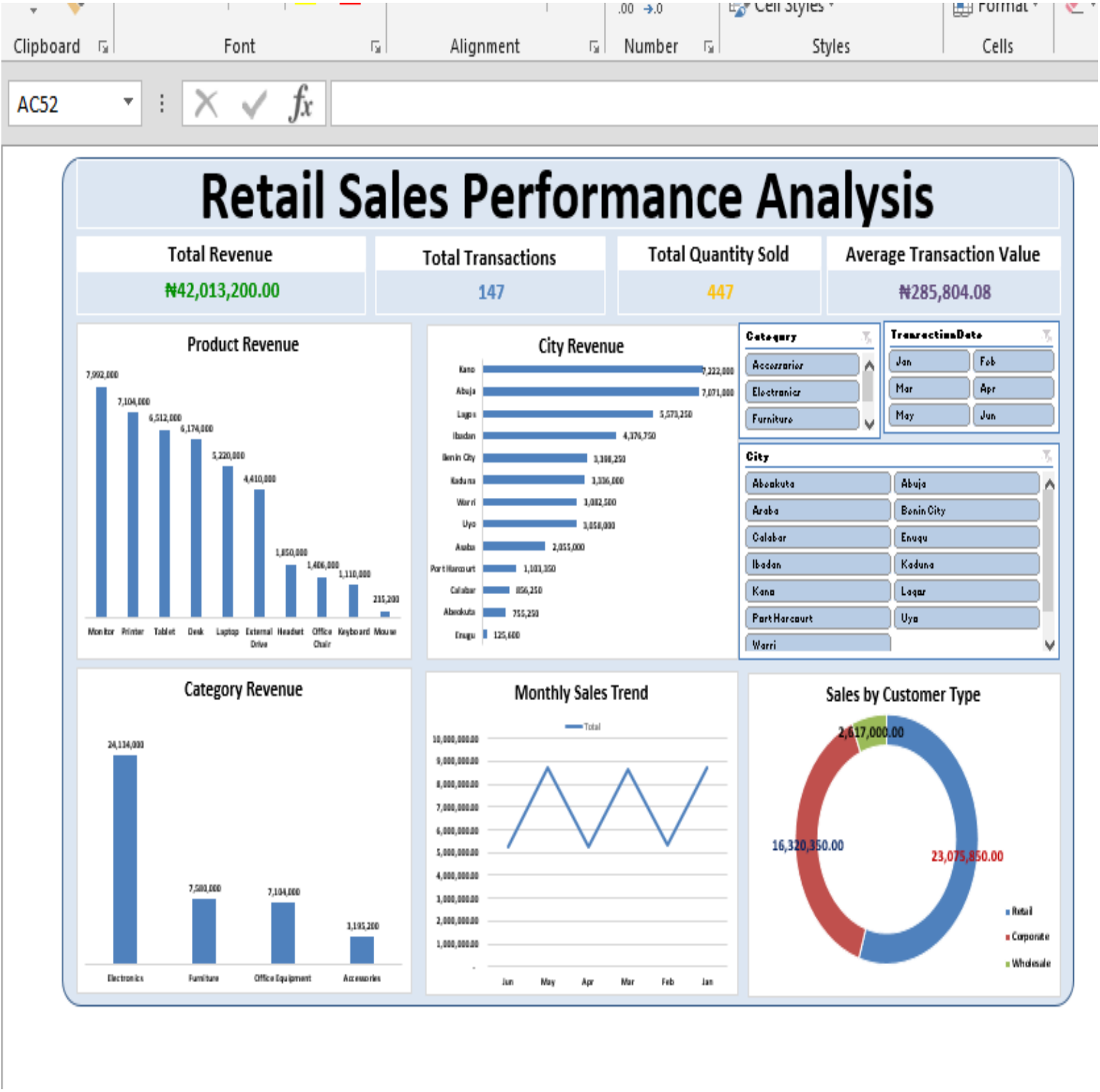
	Revenue by Customer	
	Month	**Sum of SalesAmount**
1	**Ibrahim Sani**	**4,716,000.00**
2	**Hauwa Ali**	3,768,000.00
3	**Tunde Adebayo**	3,726,000.00
4	**Mariam Yusuf**	3,454,000.00
5	**Fatima Musa**	3,336,000.00
6	Ruth Daniel	3,297,000.00
7	Esther James	3,082,500.00
8	Joy Akpan	3,058,000.00
9	Grace Eze	2,877,000.00
10	Daniel Okafor	2,355,000.00
11	David Umeh	2,055,000.00
12	Aisha Bello	1,687,500.00
13	Chinedu Obi	993,750.00
14	Michael Joseph	856,250.00
15	Peter Williams	755,250.00
16	Samuel Adeyemi	650,750.00
17	Blessing Ojo	588,750.00
18	Mercy John	521,250.00
19	Emeka Nwosu	125,600.00
20	Kelvin George	109,600.00
	Grand Total	**42,013,200.00**

Pivot Table 5 — Monthly Revenue



Monthly Revenue	
Month	**Sum of SalesAmount**
Jan	8,742,250.00
May	8,717,500.00
Mar	8,660,250.00
Feb	5,319,850.00
Apr	5,289,950.00
Jun	5,283,400.00
Grand Total	**42,013,200.00**

DASHBOARD


Clipboard
Font
Alignment
Number
Styles
Cells
AC52
Retail Sales Performance Analysis
Total Revenue
₦42,013,200.00
Total Transactions
147
Total Quantity Sold
447
Average Transaction Value
₦285,804.08
Product Revenue
7,992,000
7,104,000
6,512,000
6,174,000
5,220,000
4,410,000
1,850,000
1,406,000
1,110,000
235,200
Monitor
Printer
Tablet
Desk
Laptop
External Drive
Headset
Office Chair
Keyboard
Mouse
City Revenue
Kano
7,222,000
Abuja
7,071,000
Lagos
5,573,250
Ibadan
4,376,750
Benin City
3,388,250
Kaduna
3,336,000
Warri
3,082,500
Uyo
3,058,000
Asaba
2,055,000
Port Harcourt
1,103,350
Calabar
856,250
Abeokuta
755,250
Enugu
125,600
Category
Accessories
Electronics
Furniture
TransactionDate
Jan
Feb
Mar
Apr
May
Jun
City
Abeokuta
Abuja
Asaba
Benin City
Calabar
Enugu
Ibadan
Kaduna
Kano
Lagos
Port Harcourt
Uyo
Warri
Category Revenue
24,134,000
7,560,000
7,104,000
3,195,200
Electronics
Furniture
Office Equipment
Accessories
Monthly Sales Trend
Total
10,000,000.00
9,000,000.00
8,000,000.00
7,000,000.00
6,000,000.00
5,000,000.00
4,000,000.00
3,000,000.00
2,000,000.00
1,000,000.00
-
Jun
May
Apr
Mar
Feb
Jan
Sales by Customer Type
2,617,000.00
16,320,350.00
23,075,850.00
Retail
Corporate
Wholesale

SQL Analysis

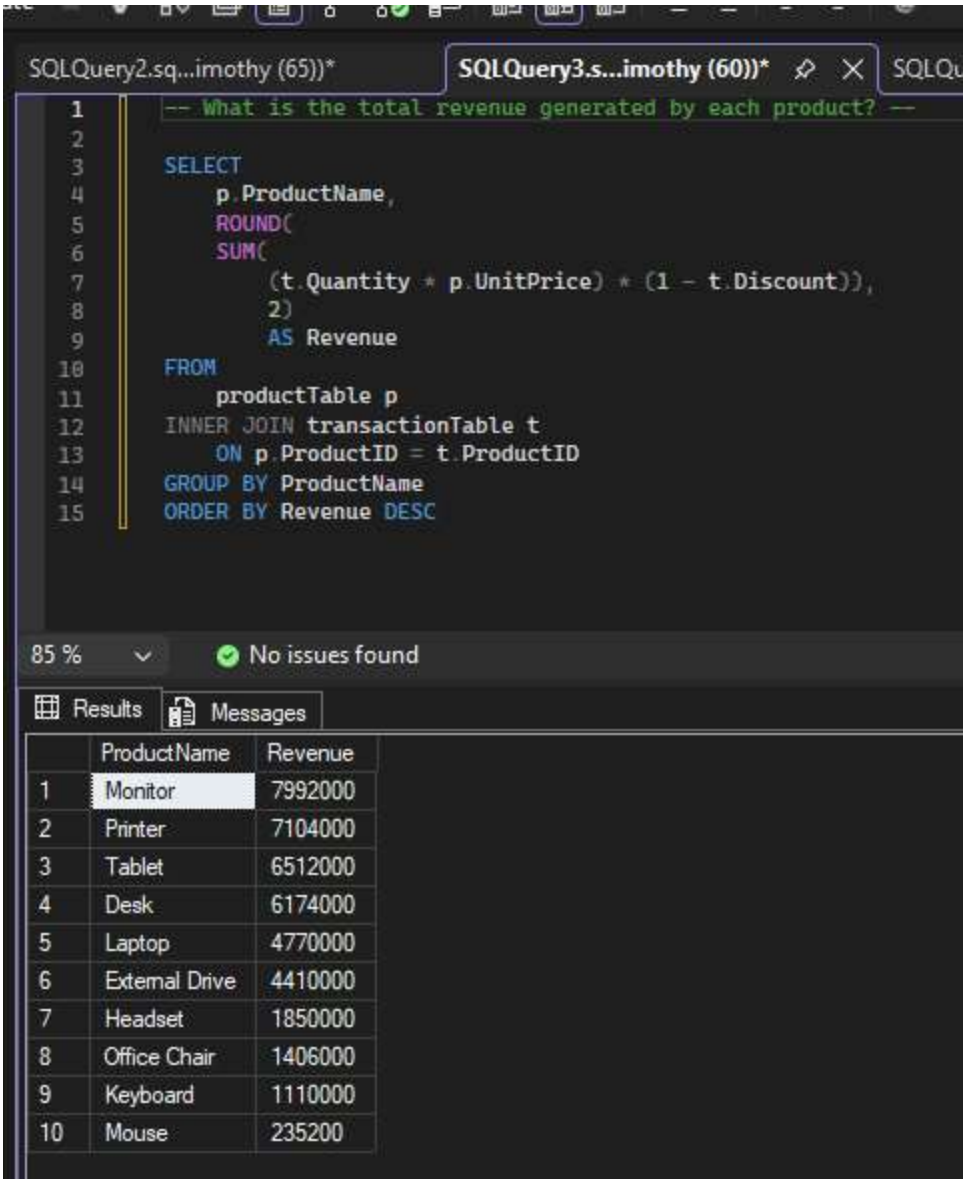
BUSINESS QUESTIONS

SQL Question 1

What is the total revenue generated by each product?

SQL QUERY:

```
SELECT
        p.ProductName,
        ROUND(SUM((t.Quantity * p.UnitPrice) * (1 - t.Discount)),
                2) AS Revenue
FROM
        productTable p
INNER JOIN transactionTable t
        ON p.ProductID = t.ProductID
GROUP BY ProductName
ORDER BY Revenue DESC
```



	ProductName	Revenue
1	Monitor	7992000
2	Printer	7104000
3	Tablet	6512000
4	Desk	6174000
5	Laptop	4770000
6	External Drive	4410000
7	Headset	1850000
8	Office Chair	1406000
9	Keyboard	1110000
10	Mouse	235200

SQL Question 2

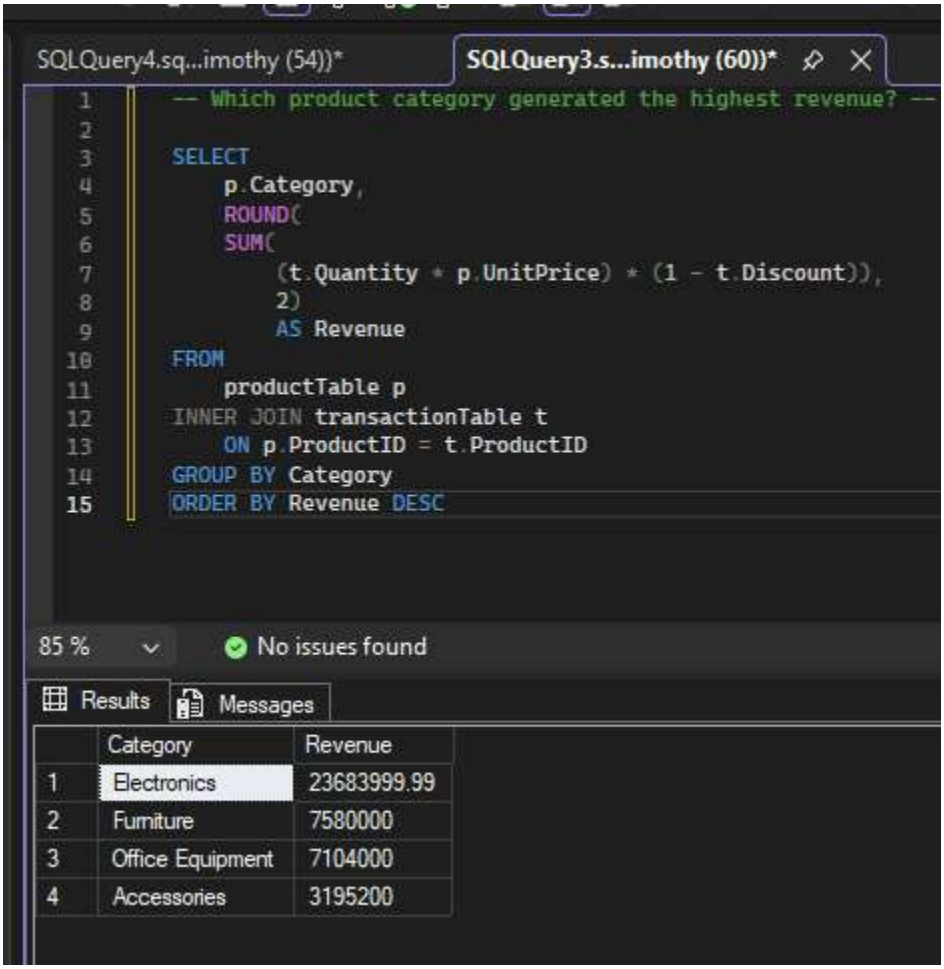
Which product category generated the highest revenue?

SQL QUERY:

```
SELECT
	p.Category,
	ROUND(
	SUM(
		(t.Quantity * p.UnitPrice) * (1 - t.Discount)),
```

```
            2)
            AS Revenue
FROM
        productTable p
INNER JOIN transactionTable t
        ON p.ProductID = t.ProductID
GROUP BY Category

ORDER BY Revenue DESC
```



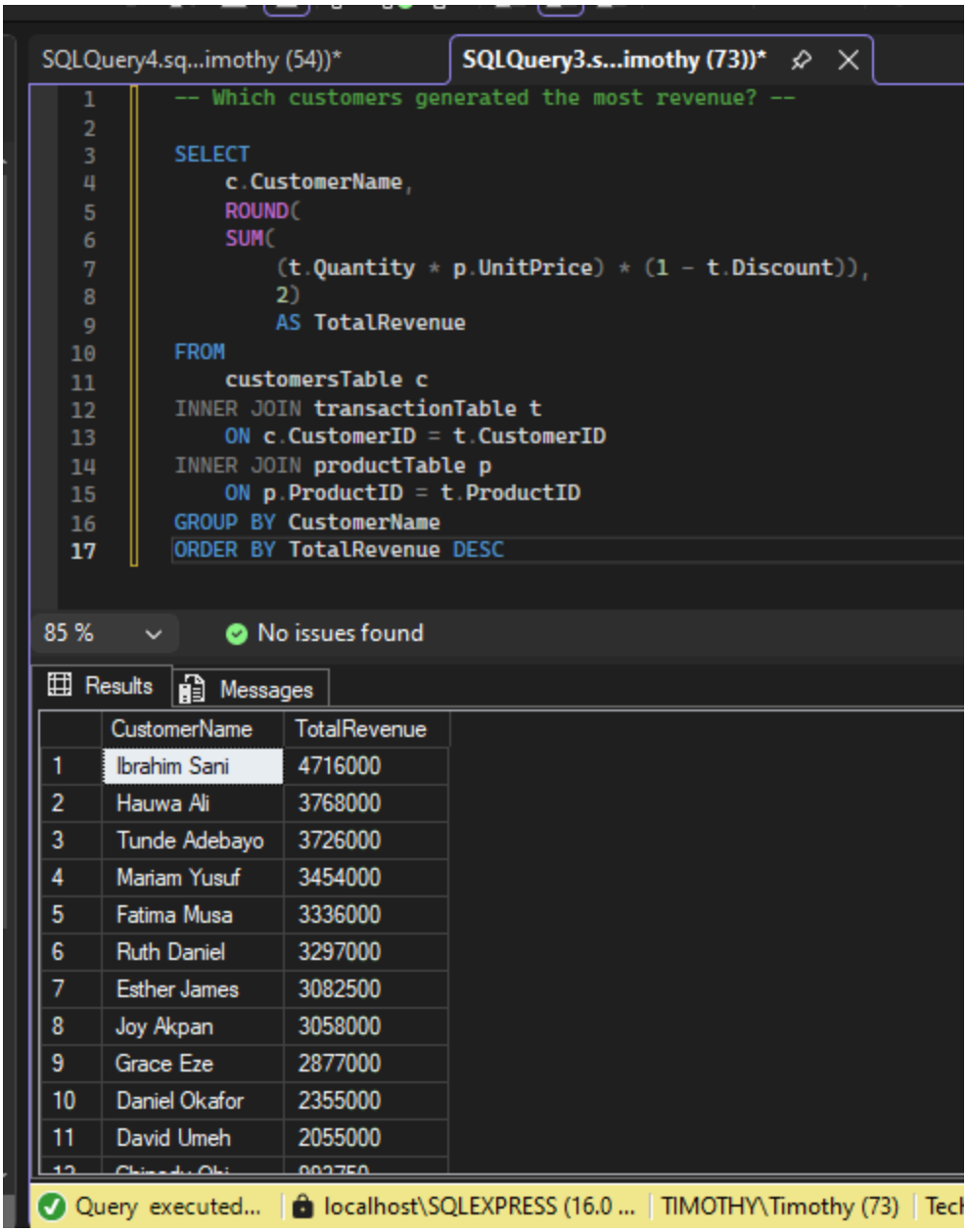
SQL Question 3

Which customers generated the most revenue?

SQL QUERY:

SELECT

```
        c.CustomerName,
        ROUND(
        SUM(
                (t.Quantity * p.UnitPrice) * (1 - t.Discount)),
                2)
                AS TotalRevenue
FROM
        customersTable c
INNER JOIN transactionTable t
        ON c.CustomerID = t.CustomerID
INNER JOIN productTable p
        ON p.ProductID = t.ProductID
GROUP BY CustomerName
ORDER BY TotalRevenue DESC
```



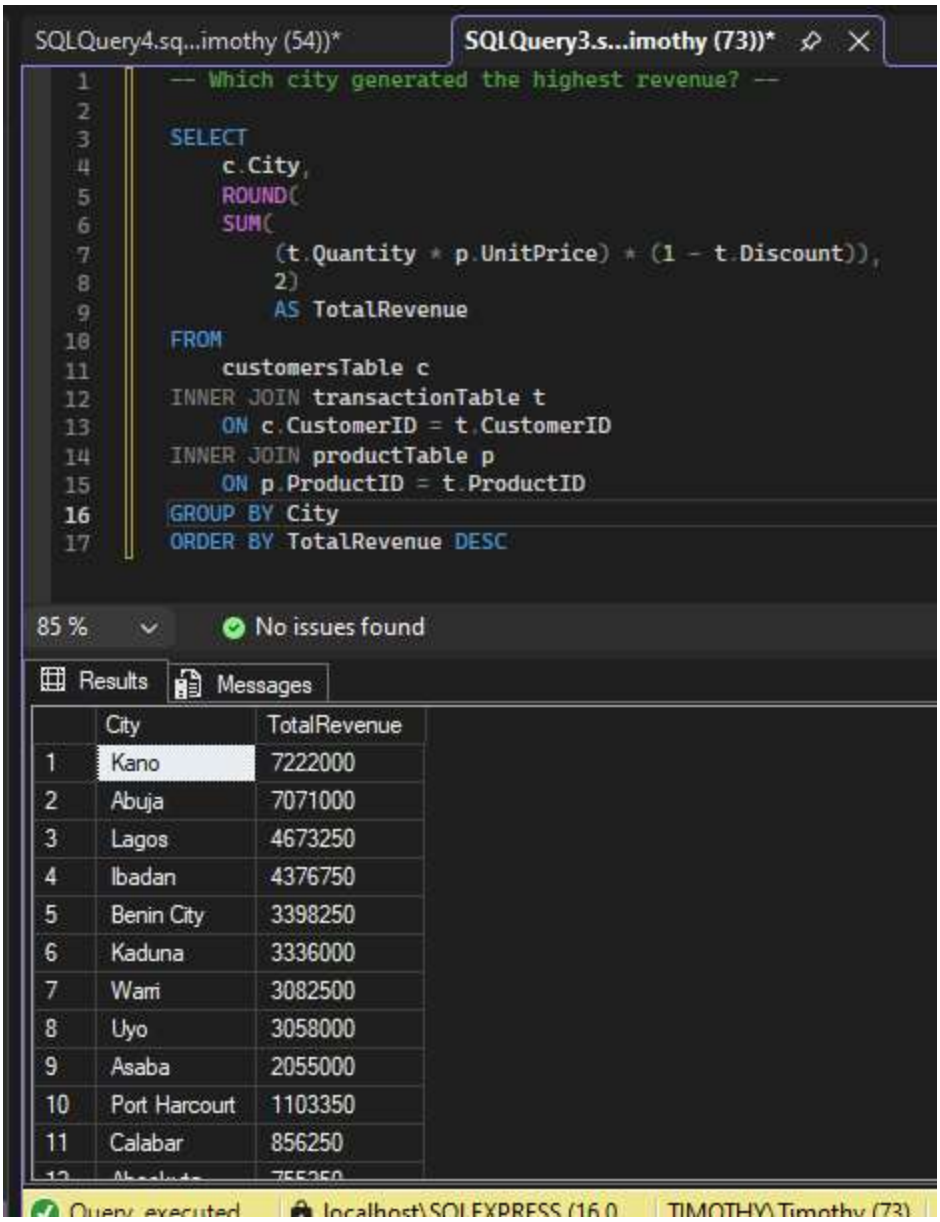
SQL Question 4

Which city generated the highest revenue?

SQL QUERY:

```
SELECT
    c.City,
```

```
        ROUND(
        SUM(
                (t.Quantity * p.UnitPrice) * (1 - t.Discount)),
                2)
                AS TotalRevenue
FROM
        customersTable c
INNER JOIN transactionTable t
        ON c.CustomerID = t.CustomerID
INNER JOIN productTable p
        ON p.ProductID = t.ProductID
GROUP BY City
ORDER BY TotalRevenue DESC
```



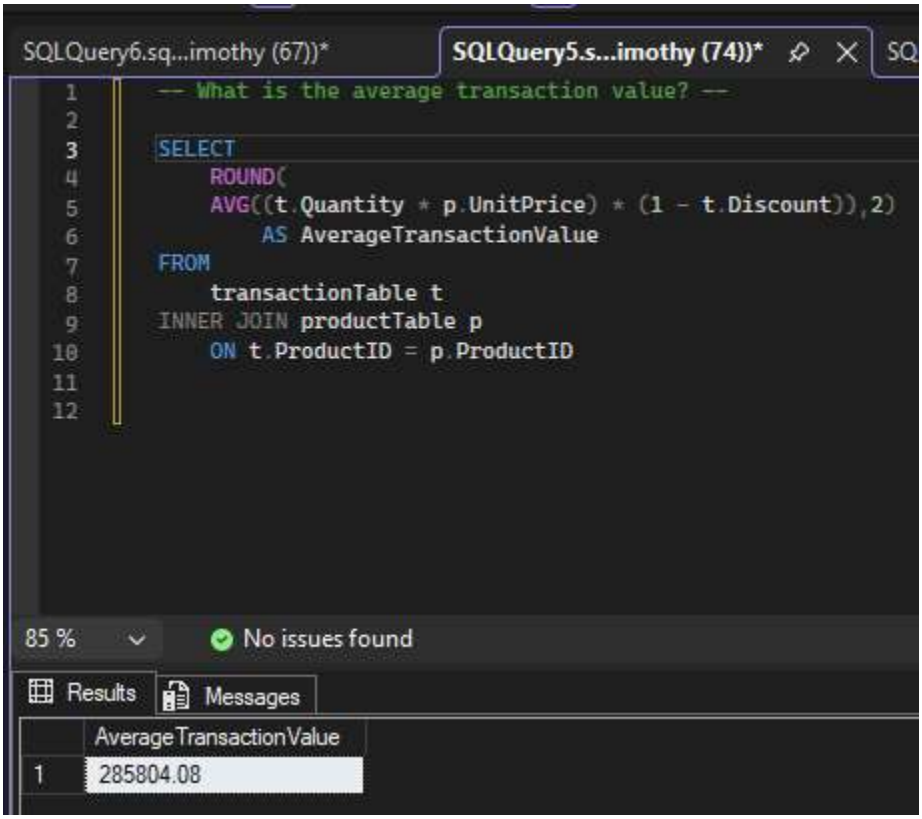
SQL Question 5

What is the average transaction value?

SQL QUERY:

```
SELECT
        ROUND(
        AVG((t.Quantity * p.UnitPrice) * (1 - t.Discount)),2)
                AS AverageTransactionValue
FROM
        transactionTable t
```

```
INNER JOIN productTable p
        ON t.ProductID = p.ProductID
```



SQL Question 6 — Optional

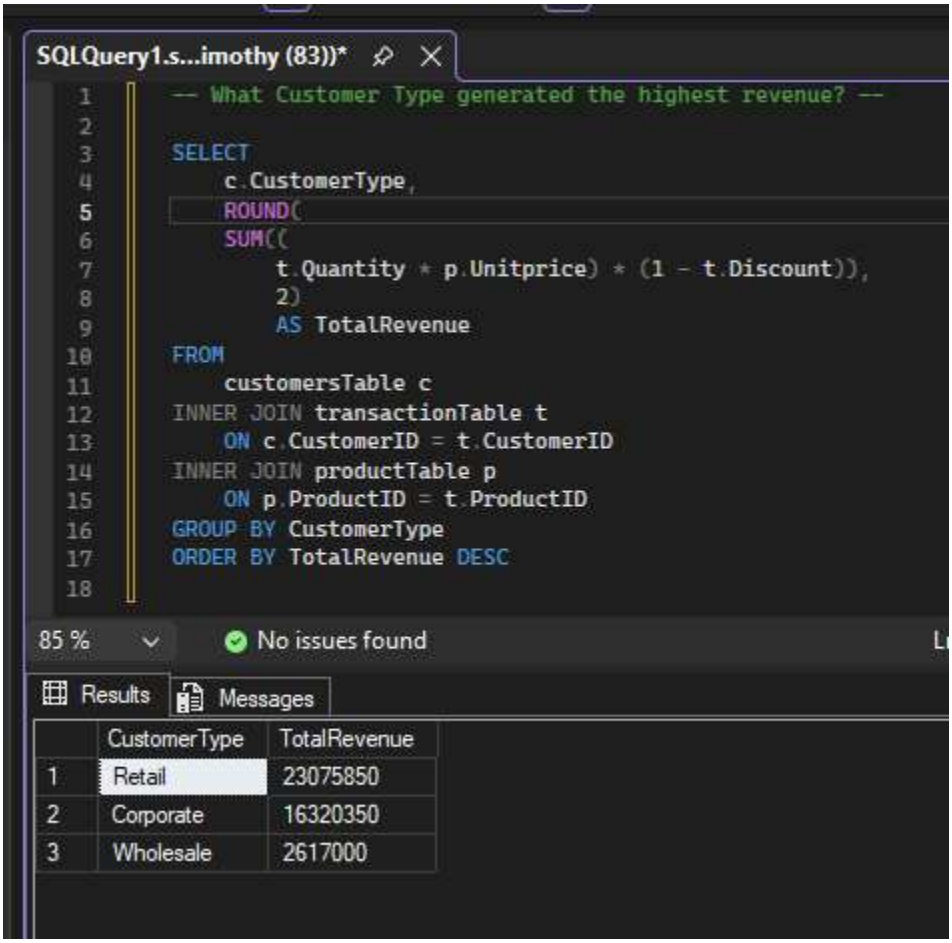
Which customer type generated the highest revenue?

SQL QUERY:

```
SELECT

        c.CustomerType,
        ROUND(
        SUM(
                (t.Quantity * p.UnitPrice) * (1 - t.Discount)),
                2)
                AS TotalRevenue
FROM
        customersTable c
INNER JOIN transactionTable t
        ON c.CustomerID = t.CustomerID
```

```
INNER JOIN productTable p
        ON p.ProductID = t.ProductID
GROUP BY CustomerType
ORDER BY TotalRevenue DESC
```



COMPARE EXCEL AND SQL RESULTS

Analysis	**Excel/Sheets Result**	**SQL Result**	**Match?**
Highest revenue product	Monitor	Monitor	Yes
Highest revenue city	Kano	Kano	Yes

Highest revenue category	Electronics	Electronics	Yes
Highest revenue by CustomerType	Retail	Retail	Yes
Highest revenue by CustomerName	Ibrahim Sani	Ibrahim Sani	Yes
Average Transaction Value	285,804.08	285,804.08	Yes

KEY INSIGHTS

Insight 1

Electronics is the dominant product category driving business revenue.
It generated **₦24,134,000.00** in revenue.
The Electronics category is the primary growth engine for sales, outperforming all other product categories (with Accessories generating the lowest at **₦3,195,200.00**).

Insight 2

Retail customers generate more revenue than any other customer segment.
Retail customers contributed **₦23,075,850.00** to total revenue.
Individual consumer purchasing represents over half of total revenue (from **₦42,013,200.00** total), making the retail segment essential to the business's bottom line.

Insight 3

Kano is the highest-performing geographic location for sales.
It generated **₦7,222,000.00** in revenue.
Kano represents the company's strongest city market and highest customer demand concentration.

BUSINESS RECOMMENDATIONS

1. **Prioritize High-Margin Electronics & Monitor Inventory in Kano**
    - **Rationale:** Electronics generated **₦24,134,000.00** (the top category), Monitors generated **₦7,992,000.00** (the top individual product), and **Kano** produced the highest city revenue at **₦7,222,000.00.**

    - **Action:** Allocate a larger share of stock and regional marketing budgets for high-performing tech items specifically to fulfillment hubs and outlets in Kano to prevent stockouts and maximize profitability in your top-performing market.

2. **Develop Tailored Retargeting & Loyalty Programs for Retail Customers and VIPs**
    - **Rationale:** Retail customers represent over **50%** of overall revenue at **₦23,075,850.00**, while top individual buyers like **Ibrahim Sani** (**₦4,716,000.00**) drive substantial value. Additionally, revenue experiences mid-year dips (dropping from a peak in January to a low point in **June**).

    - **Action:** Launch a retail loyalty program to retain individual shoppers and create targeted mid-year promotions during slower months (such as **May** and **June**) to smooth out sales fluctuations and ensure consistent revenue flow.

Conclusion

Overall, the analysis reveals a healthy retail-driven business generating **₦42,013,200.00** across **147** transactions, with high-value product categories and key markets driving the vast majority of performance. **Electronics** remains the primary engine of revenue, supported heavily by top products like **Monitors** and strong demand in regional hubs like **Kano** and **Abuja**. By strategically focusing inventory on these top-performing segments, addressing **mid-year** sales fluctuations through targeted promotions, and strengthening relationships with key retail buyers, the company can eliminate inefficiencies and drive sustainable revenue growth.

Excel worksheet

Link to Excel sheet:

https://1drv.ms/x/c/68c23cafff6a5be3/IQAXCPhD_S5iRLHuZhG5pf0NAdBqIj0OVORj0QEIfaFL10s